KEYWAVE

KW007: Ultra-Low Power 5.8GHz Microwave Radar Sensor

Motion Detection with Intelligent Doppler Engine

Programming Guide

Version 1.7

March 2026

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General Description:

The **KW007** is a **Doppler radar sensor** designed for ease of use; detection results can be **read from pin** directly, or the device can be configured and monitored via an **I²C interface**.

Once configured, the device operates autonomously, detecting object motion relative to the sensor and updating status flags to indicate **approaching** or **leaving** targets.

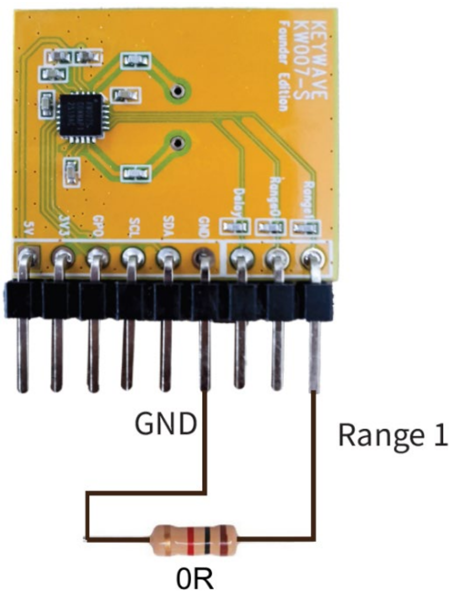
Key Operational Characteristics:

- The internal status flags (Approach/Leaving) are updated at an approximate rate of **8 Hz**.
- Hardware flags can be read directly via **I²C** for low-power **MCU** designs
- **VCO bank calibration** is factory-programmed and must be read during initialization
- Detection range is configurable through **amplifier gain** and **threshold** settings

Applications:

- **Smart Lighting:** Motion-triggered indoor/outdoor light control.
- **Touchless Control:** Hands-free faucets, soap dispensers, and automatic lids.
- **Presence Detection:** Proximity sensing for kiosks, digital signage, and smart displays.
- **Device Wake-up:** Low-power "wake-on-approach" for IoT devices and security panels.
- **Home Automation:** Context-aware sensing for smart home ecosystems.

Hardware Note



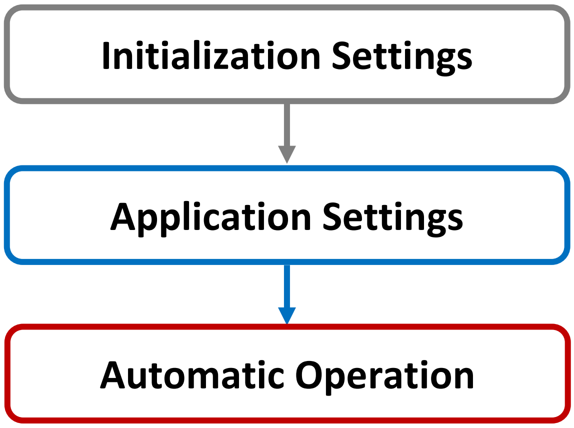
When using KW007 in **I²C mode,**
the IC pin 19 (Range1) must connect to GND.
(or pull low by MCU or 0R resistor to GND)

KW007 Radar Sensor
Official Website: https://keywavetech.co.uk/

KW007 Doppler Radar Software Procedure

The **KW007** Doppler radar sensor can operates via a standard **I²C interface**. Upon power-up, the device must perform the following initialization sequence to ensure stable operation and optimal detection performance.

1. **Initialization Settings:** Write the fundamental chip configuration parameters to initialize the device.
2. **Application Settings:** Configure the sensing parameters based on specific application requirements. This includes setting the detection range, and Doppler filter corners.
3. **Automatic Operation:** Once the configuration is applied, the radar operates automatically. The host MCU periodically reads the radar parameters at set intervals for subsequent application processing.



Standard I²C Write/Read Interface Description for KW007

Supported Modes: Standard-mode (100 kHz) / Fast-mode (400 kHz)
Device Address: 7-bit addressing
Data Format: 8-bit Data, MSB First

KW007 Write/Read Address

The KW007 uses an addressing scheme consisting of a fixed 7-bit slave address ID followed by a read/write (R/W) bit.

- **Slave address ID:** The fixed 8-bit slave address ID is **00110010b (0x32h)**.
- **Read/Write Control:** The address is followed by a R/W bit:
  - '**0**' for **Write** operations. (8bit: 00110010b, 0x32h)
  - '**1**' for **Read** operations. (8bit: 00110011b, 0x33h)

KW007 Write mode Write format is shown as follow:

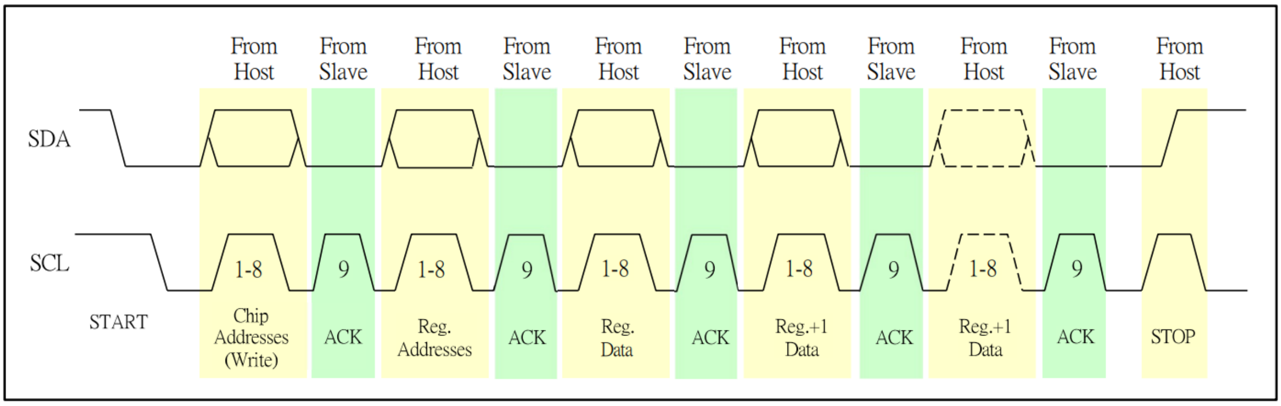


Figure 1. KW007 I²C Write Mode Timing Diagram

KW007 Read mode Read format is shown as follow:

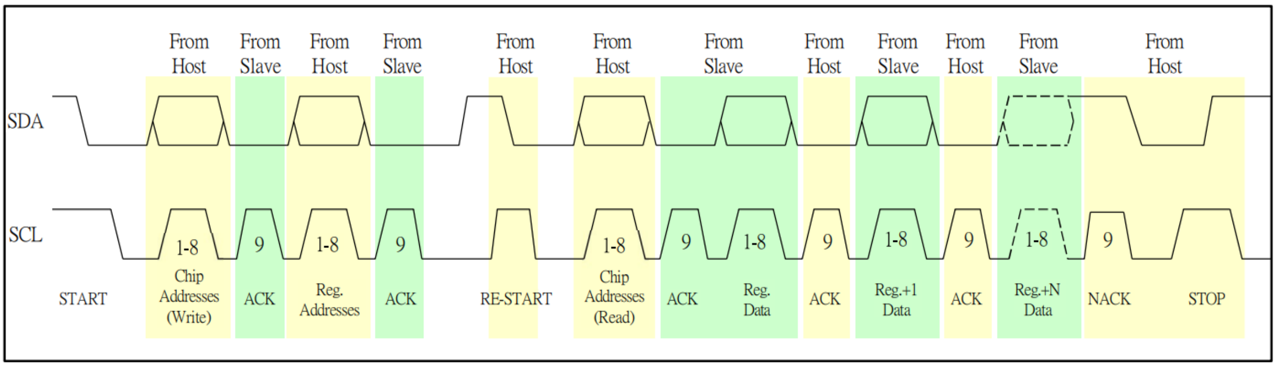


Figure 2. KW007 I²C Read Mode Timing Diagram

Register Descriptions

<table>
<tr><th>Register</th><th>Function</th><th>Access</th><th>Value</th><th>Bits</th><th colspan="2">Description</th></tr>
<tr><td>Reg 0</td><td>R0</td><td>R</td><td>-</td><td>[7:0]</td><td colspan="2">Reserved</td></tr>
<tr><td>Reg 1</td><td>VCO Bank</td><td>R</td><td>-</td><td>[5:0]</td><td colspan="2">VCO Bank (Read Only, factory burned)</td></tr>
<tr><td>Reg 2
|
Reg 4</td><td>R2
|
R4</td><td>R</td><td>-</td><td>[7:0]</td><td colspan="2">Reserved</td></tr>
<tr><td>Reg 5</td><td>RSSI</td><td>R</td><td>-</td><td>[7:0]</td><td colspan="2">RSSI Output (Read Only, 0-255)</td></tr>
<tr><td rowspan="3">Reg 6</td><td rowspan="3">Detection Flag:
Day/Night
& Approach
& Leaving</td><td rowspan="3">R</td><td rowspan="3">-</td><td>[3]</td><td colspan="2">Day/Night Flag
0: Night
1: Day</td></tr>
<tr><td>[2]</td><td colspan="2">Flag Forward/Approach
0: No object approaching
1: Object approaching</td></tr>
<tr><td>[1]</td><td colspan="2">Flag Backward/Leaving
0: No object leaving
1: Object leaving</td></tr>
<tr><td>Reg 7</td><td>R7</td><td>R</td><td>-</td><td></td><td colspan="2">Reserved</td></tr>
<tr><td>Reg 8</td><td>R8</td><td>W/R</td><td>0xFC</td><td>[7:0]</td><td colspan="2">Reserved</td></tr>
<tr><td>Reg 9</td><td>Manual Mode</td><td>W/R</td><td>0x30</td><td>[5]</td><td colspan="2">[5]: Efuse mode
0: FUSE
1: I2C</td></tr>
<tr><td>Reg 10</td><td>R10</td><td>W/R</td><td>0x88</td><td>[7:0]</td><td colspan="2">Reserved</td></tr>
<tr><td>Reg 11</td><td>R11</td><td>W/R</td><td>0x02</td><td>[7:0]</td><td colspan="2">Reserved</td></tr>
<tr><td rowspan="2">Reg 12</td><td rowspan="2">Sampling Rate</td><td>W/R</td><td rowspan="2">0x54</td><td>[7]</td><td colspan="2">Sampling Rate: 0=1kHz, 1=500Hz</td></tr>
<tr><td>W/R</td><td>[5:0]</td><td colspan="2">VCO Bank (Set during initialization)</td></tr>
<tr><td>Reg 13</td><td>R13</td><td>W/R</td><td>0x4B</td><td>[1]</td><td colspan="2">Set 16kHz clk to Delay output.
0: Set clk to output
1: Off</td></tr>
<tr><td>Reg 14
|
Reg 18</td><td>R14
|
R18</td><td>W/R</td><td>-</td><td>[7:0]</td><td colspan="2">Reserved</td></tr>
<tr><td>Reg 19</td><td>Amp1_Gain</td><td>W/R</td><td>0xEA</td><td>[7:5]</td><td>Amplifier 1 Gain:
000: 8 db
001: 12 db
010: 16 db
011: 19 db
100: 21 db
101: 24 db
110: 28 db
111: 32 db</td><td rowspan="2">Total Gain =
Amp1_Gain
+
Amp2_Gain
+
Amp3_Gain

(Ex. 24+24+0 = 48db)</td></tr>
<tr><td>Reg 20</td><td>Amp3_Gain</td><td>W/R</td><td>0xF7</td><td>[4:3]</td><td>Amplifier 3 Gain:
00: 0 db
01: 4 db
10: 8 db
11: 12 db</td></tr>
</table>

Register	Function	Access	Value	Bits	Description
	Amp2_Gain	W/R		[2:0]	Amplifier 2 Gain: 000: 8 db 001: 12 db 010: 16 db 011: 19 db 100: 21 db 101: 24 db 110: 28 db 111: 32 db
Reg 21	Slow Speed Detection	W/R	0x5F	[5]	High-pass Corner: (Default = 0) 0: Low (Allows slow-moving signals.) 1: High (Cuts off slow-moving signals.)
Reg 22 | Reg 23	R22 | R23	W/R	-	[7:0]	Reserved
Reg 24	Enhanced Mode	W/R	0xA0	[6]	Enhanced Range Mode: 0: Off (Normal mode) 1: On (Enhanced mode)
Reg 25 | Reg 26	R25 | R26	W/R	-	[7:0]	Reserved
Reg 27	Valid Decision Magnitude Threshold	W/R	0x30	[7:0]	Decision Magnitude Threshold 00000000: 0 00000001: 1 00000010: 2 … 11111111: 255
Reg 28	R28	W/R	0x55	[7:0]	Reserved
Reg 29	GPO Delay	W/R	0x65	[2:1]	GPO Delay: (Default = 11) 00: 16s 01: 32s 10: 1s 11: 64s
	High Speed Detection	W/R		[0]	Low-pass Corner: (Default = 1) 0: Low (Filters out ultra-fast signals.) 1: High (allow more fast objects detect.)
Reg 30 | Reg 31	R30 | R31	W/R	-	[7:0]	Reserved

KW007 Recommended Register Configuration:

Address	Value (Hex)	Address	Value (Hex)	Address	Value (Hex)	Address	Value (Hex)
Reg 0	0x00	Reg 8	0xFC	Reg 16	0x98	Reg 24	0xA0
Reg 1	0x00	Reg 9	0x30	Reg 17	0xA0	Reg 25	0x20
Reg 2	0x00	Reg 10	0x88	Reg 18	0x00	Reg 26	0x9A
Reg 3	0x00	Reg 11	0x03	Reg 19	0xEA	Reg 27	0x30
Reg 4	0x00	Reg 12	0x54	Reg 20	0xF7	Reg 28	0x55
Reg 5	0x00	Reg 13	0x4B	Reg 21	0x5F	Reg 29	0x75
Reg 6	0x00	Reg 14	0x68	Reg 22	0x02	Reg 30	0x42
Reg 7	0x00	Reg 15	0x55	Reg 23	0x30	Reg 31	0x00

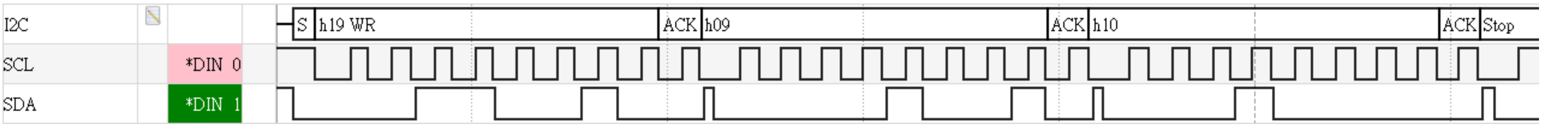
Note: Reg9 = 0x30 (manual mode), Reg12[5:0] will be set to VCO bank value.

Initialization Flow

Follow these steps in order:

1. **Write REG9 = 0x10** (Efuse mode) to enable reading the factory-burned VCO bank value
2. **Read REG1[5:0]** to obtain the VCO bank value (factory calibrated for each chip)
3. **Set REG12[5:0] = VCO bank value** in your initialization register array
4. **Write all 32 registers (R0-R31)** with REG9 = 0x30 (I²C manual mode)

Write Reg9 = 0x10



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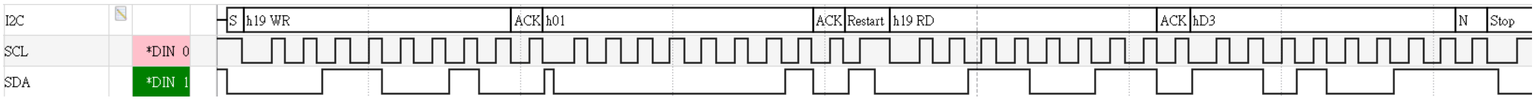
Read Reg1[5:0]



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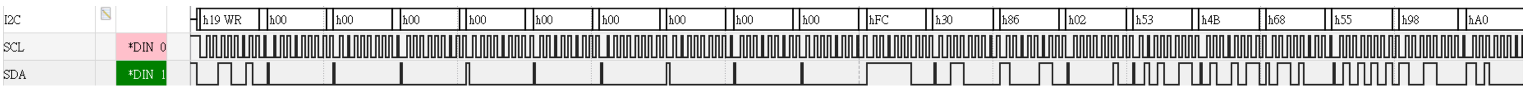
Set Reg12[5:0] = VCO bank

↓

Write all 32 Registers



1. Functional Description

The detection range is determined by the three-stage amplifier gain (AmpX_Gain), the decision magnitude threshold. Increasing the amplifier gain and reducing the magnitude threshold will extend the detection range; however, these settings also increase susceptibility to false triggers.

Three-stage Amplifier Gain:

Amp1_Gain R19[7:5]:

(000=8dB, 001=12dB, 010=16dB, 011=19dB, 100=21dB, 101=24dB, 110=28dB, 111=32dB)

Amp2_Gain R20[2:0]:

(000=8dB, 001=12dB, 010=16dB, 011=19dB, 100=21dB, 101=24dB, 110=28dB, 111=32dB)

Amp3_Gain R20[4:3]:

(000=0dB, 001=4dB, 010=8dB, 011=12dB)

For example, to set gain to 48 dB:

Write amp1_gain 32dB, write amp2_gain 16dB, write amp3_gain 0dB.

Total gain:

32 + 16 + 0 = 48 dB

(We suggested to use higher gain in previous stage for better signal/noise quality.)

Mag Threshold:

Reg27 set the signal threshold after three-stage amplifier, signal need larger than mag threshold to do detection algorithm. (Range 0~255).

Note: If gain too small or too high, the signal strength saturated to 0 or 255 and algorithm not work well.

Enhance Range Mode:

When Reg24[6] = 1, the detection range will increase.

Doppler Filter High-Pass Corner

The Reg21[5] defines the high-pass corner frequency of the Doppler filter.

- 0: Low corner frequency.
  Allows low-frequency signals to pass; capable of detecting subtle movements (e.g., breathing).
- 1: High corner frequency.
  Cuts off slow-moving signals; optimized for detecting rapid motion and filtering out environmental drifts.

Doppler Filter Low-Pass Corner

The Reg29[0] defines the low-pass corner frequency of the Doppler filter.

- 0: Low corner frequency.
  Filters out ultra-fast signals (likely interference) to provide a steady and reliable output.
- 1: High corner frequency.
  Best for speed. It allows the radar to "see" everything instantly, making it highly sensitive to objects moving at extreme speeds.

Sampling Rate:

Sampling rate Reg12[7] will influence filter corner.

Ex. Corner,fs(1kHz) = 2 x Corner,fs(500Hz).

*fs = Sampling Frequency.

The detection object speed is determined by doppler filter **corner** and **sampling rate**.
The Doppler low-pass corner filters out fast-moving clutter, but may also attenuate fast-walking human targets. The Doppler high-pass corner is related to slow-motion clutter rejection.

GPO Delay Time

The Reg29[2:1] defines the delay time for the GPO output.

To prevent the output from turning ON and OFF too quickly (chattering) when a target is at the edge of the detection range.

Once motion is detected, the GPO triggers. When the motion stops, the GPO will wait for the **Delay Time** to expire before returning to the "Idle" state.

- 00: 16 Sec
- 01: 32 Sec
- 10: 1 Sec
- 11: 64 Sec

Approach Flag

The Reg6[2] defines whether an object is moving toward the radar.
Read only.

- 0: No object approaching.
- 1: Object approaching.

Leaving Flag

The Reg6[1] defines whether an object is moving away from the radar.
Read only.

- 0: No object leaving.
- 1: Object leaving.

Day/Night Flag

The Reg6[1] defines ambient light condition.
Read only.

- 0: Night.
- 1: Day.

Recommended Settings Table

Detection Range (Typical Estimate)	**Total Gain** Reg19[7:5] + Reg20[4:0] = AMP1 + AMP2 + AMP3	**Magnitude Threshold** Reg27[7:0]	**Enhanced Mode** Reg24[6]
Level 0 (15 cm)	**19+8+0 = 27 dB**	**45**	**OFF**
Level 1 (50cm)	**24+8+0 = 32dB**	**64**	**OFF**
Level 2 (100 cm)	**32+12+0 = 44 dB**	**90**	**OFF**
Level 3 (3m)	**32+21+0 = 53 dB**	**90**	**OFF**
Level 4 (5 m)	**32+32+0 = 64 dB**	**90**	**OFF**
Level 5 (7.5 m)	**32+32+4 = 68 dB**	**64**	**OFF**
Level 6 (10 m)	**32+32+8 = 72 dB**	**32**	**OFF**
Level 7 (13 m)	**32+32+8 = 72 dB**	**32**	**ON**

NOTICE & CONFIDENTIALITY STATEMENT

Confidentiality and Proprietary Rights

Revision History	
V1.6	Reg29 0x65 -> 0x75
V1.7	Modify Reg9 Description Update Recommended Table (Level 0~7)

End of Document